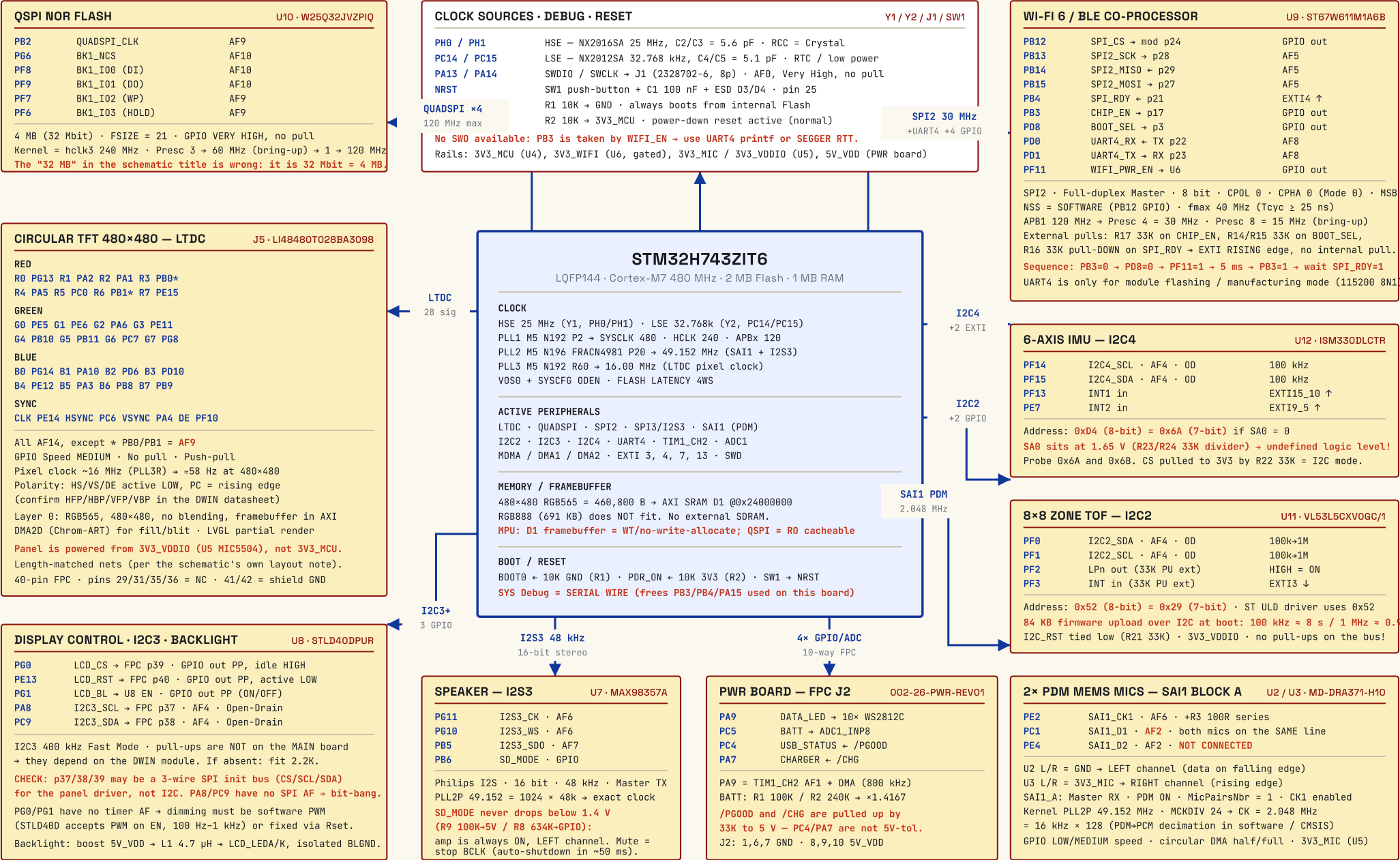


STM32H743ZIT6 — BLOCK DIAGRAM & PIN MAP



COMPLETE PIN MAP — BY PORT

PORT A16 used			
PIN	#	NET	FUNCTION / MODE
PA1	35	LCD_R2	LTDC AF14
PA2	36	LCD_R1	LTDC AF14
PA3	37	LCD_B5	LTDC AF14
PA4	40	LCD_VSYNC	LTDC AF14
PA5	41	LCD_R4	LTDC AF14
PA6	42	LCD_G2	LTDC AF14
PA7	43	CHARGER	GPIO_IN, no pull
PA8	100	LCD_SCL	I2C3_SCL AF4 0D
PA9	101	DATA_LED	TIM1_CH2 AF1
PA10	102	LCD_B1	LTDC AF14
PA13	105	SWDIO	SYS AF0
PA14	109	SWCLK	SYS AF0

PORT B15 used			
PIN	#	NET	FUNCTION / MODE
PB0	46	LCD_R3	LTDC AF9
PB1	47	LCD_R6	LTDC AF9
PB2	48	QSPI_CLK	QUADSPI AF9
PB3	133	WIFI_EN	GPIO_OUT (ex-JTD0)
PB4	134	WIFI_IRQ	EXTI4 ↑ (ex-NJTRST)
PB5	135	I2S_DATA	I2S3_SDO AF7
PB6	136	I2S_SDMODE	GPIO_OUT PP
PB8	139	LCD_B6	LTDC AF14
PB9	140	LCD_B7	LTDC AF14
PB10	69	LCD_G4	LTDC AF14
PB11	70	LCD_G5	LTDC AF14
PB12	73	WIFI_NSS	GPIO_OUT PP (SW NSS)
PB13	74	WIFI_SCK	SPI2_SCK AF5
PB14	75	WIFI_MISO	SPI2_MISO AF5
PB15	76	WIFI_MOSI	SPI2_MOSI AF5

PORT C9 used			
PIN	#	NET	FUNCTION / MODE
PC0	26	LCD_R5	LTDC AF14
PC1	27	SAI_D1	SAI1_D1 AF2
PC4	44	USB_STATUS	GPIO_IN, no pull
PC5	45	BATT	ADC1_INP8 ANALOG
PC6	96	LCD_HSYNC	LTDC AF14
PC7	97	LCD_G6	LTDC AF14
PC9	99	LCD_SDA	I2C3_SDA AF4 0D
PC14	8	OSC32_IN	RCC LSE

PORT D5 used			
PIN	#	NET	FUNCTION / MODE
PD0	114	WIFI_TX	UART4_RX AF8
PD1	115	WIFI_RX	UART4_TX AF8
PD6	122	LCD_B2	LTDC AF14
PD8	77	WIFI_BOOT	GPIO_OUT PP
PD10	79	LCD_B3	LTDC AF14

PORT E10 used			
PIN	#	NET	FUNCTION / MODE
PE2	1	SAI_CK	SAI1_CK1 AF6
PE4	3	SAI_D2	dead net — Leave unused
PE5	4	LCD_G0	LTDC AF14
PE6	5	LCD_G1	LTDC AF14
PE7	58	IMU_INT2	EXTI9_5 ↑
PE11	64	LCD_G3	LTDC AF14
PE12	65	LCD_B4	LTDC AF14
PE13	66	LCD_RST	GPIO_OUT PP
PE14	67	LCD_CLK	LTDC AF14
PE15	68	LCD_R7	LTDC AF14

PORT F13 used			
PIN	#	NET	FUNCTION / MODE
PF0	10	ToF_SDA	I2C2_SDA AF4 0D
PF1	11	ToF_SCL	I2C2_SCL AF4 0D
PF2	12	ToF_LPn	GPIO_OUT PP
PF3	13	ToF_INT	EXTI3 ↓
PF6	18	QSPI_I03	BK1_I03 AF9
PF7	19	QSPI_I02	BK1_I02 AF9
PF8	20	QSPI_I00	BK1_I00 AF10
PF9	21	QSPI_I01	BK1_I01 AF10
PF10	22	LCD_DE	LTDC AF14
PF11	49	WIFI_PWR_EN	GPIO_OUT PP
PF13	53	IMU_INT1	EXTI15_10 ↑
PF14	54	IMU_SCL	I2C4_SCL AF4 0D
PF15	55	IMU_SDA	I2C4_SDA AF4 0D

PORT G / H / SYS10 used			
PIN	#	NET	FUNCTION / MODE
PG0	56	LCD_CS	GPIO_OUT PP
PG1	57	LCD_BL	GPIO_OUT PP
PG6	91	QSPI_NCS	BK1_NCS AF10

FREE PINS (NC IN THE SCHEMATIC)41	
PA0 · PA11 · PA12 · PA15 · PB7 · PC2 · PC3 · PC8 · PC10 · PC11 · PC12 · PC13 · PD2 · PD3 · PD4 · PD5 · PD7 · PD9 · PD11 · PD12 · PD13 · PD14 · PD15 · PE0 · PE1 · PE3 · PE8 · PE9 · PE10 · PF4 · PF5 · PF12 · PG2 · PG3 · PG4 · PG5 · PG7 · PG9 · PG12 · PG15 · PE4	
- Set all of them to Analog in CubeMX — lowest leakage and least coupling. - PA11/PA12 = USB_OTG_FS, but the USB-C sits on the PWR board and never reaches the MCU. Do not enable USB. - PC13/PC2/PC3 are free for expansion; PD8/PD9 would give you USART3 if you want a dedicated console.	

GPIO SPEED — DECISION TABLEVDD 3.3 V, CL 30 pF			
GROUP	SPEED	PULL	TYPE
QSPI ×6	VERY HIGH	None	AF PP
SPI2 ×3 + NSS	VERY HIGH	None	AF PP
LTDC ×28	MEDIUM	None	AF PP
DATA_LED (PA9)	HIGH	None	AF PP
SAI1 CK/D1	LOW	None	AF PP
I2S3 ×3	LOW	None	AF PP
I2C ×6 (all)	LOW	Pull-up*	AF 0D
SWDIO / SWCLK	VERY HIGH	None	AF PP
Slow GPIO outputs	LOW	None	OUT PP
EXTI / status inputs	—	None**	INPUT
* Internal pull-up is only a stopgap — see Sheet 4. ** Every EXTI and status line already has an external 33K resistor; do not enable an internal pull that would fight it.			

EXTI — SUMMARY4 lines · NVIC				
PIN	SIGNAL	EDGE	VECTOR	PRI0
PB4	WIFI_RDY	Rising	EXTI4_IRQn	5
PF3	ToF_INT	Falling	EXTI3_IRQn	7
PF13	IMU_INT1	Rising	EXTI15_10	6
PE7	IMU_INT2	Rising	EXTI9_5	6
PB4 is NJTRST and comes out of reset with an internal pull-up. Together with the external 33K pull-down the level is undefined until the GPIO is configured — configure PB4 early, before enabling the module. - Audio DMA (SAI/I2S) must have higher NVIC priority (lower number) than any EXTI, otherwise you get audible glitches. - PC4/PA7 (charger status) don't need EXTI: poll every 100 ms with debounce.				

PER-PERIPHERAL CONFIGURATION FOR STM32CUBEMX

RCC • CLOCK TREE		HSE 25 MHz
HSE	Crystal/Ceramic Resonator, 25 MHz	
LSE	Crystal/Ceramic Resonator, 32.768 kHz	
PLL1	M=5, N=192, P=2, Q=4, R=2 → 480 MHz VCI range 4-8 MHz • VCO wide	
D1CPRE	/1 → CPU 480 MHz	
HPRE	/2 → HCLK / AXI / AHB = 240 MHz	
APB1-4	/2 → 120 MHz (timers run at 240 MHz)	
PLL2	M=5, N=196, FRACN=4981, P=20 → 49.152 MHz (audio, exact)	
PLL3	M=5, N=192, R=60 → 16.00 MHz (LTDC)	
SAI1 clk	PLL2P	
SPI1/2/3	PLL2P (exact I2S3 for 48 kHz)	
QSPI clk	rcc_hclk3 (240 MHz)	
I2C1/2/3	PCLK1 • I2C4 → PCLK4	
ADC clk	PLL2P / async, target ≤ 25 MHz	
MANDATORY FOR 480 MHZ Power Regulator Voltage Scale = VOSO and enable SYSCFG_PWRCR_ODEN before raising the PLL. Flash latency = 4 WS. Without this the MCU hangs inside SystemClock_Config().		

LTDC		RGB888 pins · 480×480
Pixel format	Layer0 = RGB565 (saves RAM)	
Pixel clock	16.00 MHz (PLL3R)	
HSYNC / VSYNC	Active LOW	
DE	Active LOW	
PCLK polarity	Rising edge (input pixel clock)	
Starting point	HSW 8 · HBP 20 · HFP 20	
	VSW 4 · VBP 16 · VFP 16	
Framebuffer	0x24000000 (AXI SRAM D1, 512 KB)	
Background	0x000000	
- Those blanking numbers are a starting point: replace them with the DWIN datasheet values before you freeze the firmware. - Enable DMA2D (Chrom-ART) — without it any full-screen fill at 480×480 eats the CPU. - MPU: mark the framebuffer region <i>Write-Through, no write allocate</i>; skip this and you'll see tearing/artifacts with D-Cache on.		

QUADSPI		W25Q32JV · 4 MB
Mode	Bank 1 · Single Flash · 4 lines	
Prescaler	3 (±4 → 60 MHz) for bring-up	
	1 (±2 → 120 MHz) once validated	
Fifo threshold	4	
Sample shifting	Sample M1 SPI settings: Half cycle (at ≥ 60 MHz) ST67W61 (POL 0, PHA 0, 8 bit, MSB, max 40 MHz)	

SPI2 — ST67W611M1

Wi-Fi/BLE NCP

Mode	Full-Duplex Master
NSS	Software (PB12 as plain GPIO)
Data size	8 bits
First bit	MSB First
CPOL	Low (0)
CPHA	1 Edge (0)
Prescaler	8 → 15 MHz (bring-up) 4 → 30 MHz (production)
Module fmax	40 MHz (Tcyc ≥ 25 ns)
CRC	Disabled
DMA	TX + RX, Normal, byte

Handshake (module protocol)

- Host wants to send → raise CS (PB12) → wait for SPI_RDY (PB4) = 1 → start the clock → transfer.
- Module wants to send → it raises SPI_RDY on its own → EXTI4 → host raises CS and clocks the data out.
- Only lower CS after SPI_RDY has gone low. Dropping it early puts the module into an error state.
- Frame = 8-byte header (sync 0x55AA, len, frame, type, reserved) + payload that is a multiple of 4 (pad with 0x88).

Use the X-CUBE-ST67W61 package: it already ships the AT/SPI driver, the handshake and LwIP. Just point the handles at SPI2 / PB12 / PB4 / PB3 / PD8.

SAI1 BLOCK A — 2 PDM MEMS		MD-DRA371-H10
Mode	Master Receive	
PDM	Enabled · MicPairsNbr = 1	
Clock enable	CK1 only (CK2 off)	
Protocol	Free protocol	
Data size	16 bits · MSB first	
Slots	2 slots × 16 bit, frame 32	
Kernel	PLL2P 49.152 MHz	
MCKDIV	24 → CK1 = 2.048 MHz	
MCLK out	Disabled	
DMA	DMA1 Stream · Circular · half-word	
2.048 MHz = 16 kHz × 128 (decimation 128). For 8 kHz use MCKDIV 48; for 32 kHz use 12. - Both mics share PC1: U2 (L/R=GND) drives on the falling edge, U3 (L/R=VDD) on the rising edge. The SAI splits them into the 2 slots for you. - PDM→PCM decimation: ST's PDM2PCM library (CMSIS DSP) or the X-CUBE-AUDIO filter. - PE4/SAI_D2 is routed at the MCU but reaches no microphone — do not configure it.		

I2C — THE THREE BUSES			All independent	
BUS	PINS	DEVICE	ADDR 7B	SPEED
I2C2	PF0/PF1	VL53L5CX	0x29	100k*
I2C3	PC9/PA8	DWIN TFT	—	400k
I2C4	PF15/PF14	ISM330DLC	0x6A*	100k*
Common CubeMX settings				
Mode	I2C			
Speed Mode	Standard 100 kHz (see note)			
Rise time	400 ns (SM) / 100 ns (FM)			
Fall time	100 ns (SM) / 50 ns (FM)			
Analog filter	Enabled			
Digital filter	0			
Own address	0 · 7-bit · No stretch = Disabled			
GPIO	AF4 · Open-Drain · Pull-up · LOW			

NO PULL-UPS ON THE PCB The schematic has **no pull-up resistors** on any of the three I2C buses. You *can* get going on the internal pull-up (~40 kΩ), but that caps you at 100 kHz and is fragile. **Fit 2.2 kΩ on SDA and SCL of every bus** — otherwise the VL53L5CX alone takes ~8 s just to load its firmware at boot.

8-bit addresses, as printed on the schematic: ToF 0x52 (ST's ULD driver uses this form), IMU 0xD4.

TIM1_CH2 + DMA — WS2812C ×10		PA9 · PWR board
Timer clock	240 MHz (APB2 ×2)	
Prescaler	0	
Counter Period	299 → 800 kHz (1.25 μs)	
Channel 2	PWM Generation CH2 · AF1	
CCR for bit "0"	96 (0.40 μs)	
CCR for bit "1"	192 (0.80 μs)	
Reset latch	≥ 300 μs held low	
DMA	DMA2 · Mem→Periph · Normal · half-word	
DMA target	TIM1→CCR2	
Buffer	10 × 24 + 240 zeros = 480 half-words	
Bit order	G R B, MSB first, per LED	
MARGINAL LOGIC LEVEL The WS2812Cs run off 5V_LED and need $V_{IH} \approx 0.7 \times 5\text{ V} = \mathbf{3.5\text{ V}}$, but PA9 only delivers 3.3 V. R14 (330 Ω series) doesn't fix that. If LED1 shows the wrong colour: add a level shifter, drop the LED rail to ~4.3 V with a series diode, or sacrifice LED1 as a buffer.		

ADC1 — BATTERY SENSE		PC5 · through the FPC
Channel	ADC1_INP8 (PC5), single-ended	
Resolution	16 bits	
Sampling time	810.5 cycles (the maximum)	

GPIOs, BOOT SEQUENCE AND HARDWARE GOTCHAS

EVERY DISCRETE GPIO required initial state

PIN	SIGNAL	MODE	INIT	ACTIVE
PF11	WIFI_PWR_EN	OUT PP	0	HIGH = LDO ON
PB3	WIFI_EN	OUT PP	0	HIGH = chip ON
PD8	WIFI_BOOT	OUT PP	0	LOW = mission mode
PB12	WIFI_NSS	OUT PP	0	HIGH = selected
PB4	WIFI_RDY	IN EXTI↑	–	HIGH = ready
PE13	LCD_RST	OUT PP	0	LOW = in reset
PG0	LCD_CS	OUT PP	1	LOW = asserted
PG1	LCD_BL	OUT PP	0	HIGH = light ON
PF2	ToF_LPn	OUT PP	1	HIGH = I2C alive
PF3	ToF_INT	IN EXTI↓	–	LOW = data ready
PF13	IMU_INT1	IN EXTI↑	–	HIGH = event
PE7	IMU_INT2	IN EXTI↑	–	HIGH = event
PB6	I2S_SDMODE	OUT PP	1	no real effect
PA9	DATA_LED	AF1 PWM	0	800 kHz NRZ
PC4	USB_STATUS	IN no pull	–	LOW = USB OK
PA7	CHARGER	IN no pull	–	LOW = charging
PC5	BATT	ANALOG	–	ADC1_INP8

PWR BOARD INTERFACE BQ24075 + TPS61022

- **USB_STATUS** = /PGOOD, open-drain. **LOW** = 5 V USB present and in range; HIGH = no USB.
- **CHARGER** = /CHG, open-drain. **LOW** = charging; HIGH = done, suspended, or no source.
- States: PG00D=0, CHG=0 charging · PG00D=0, CHG=1 complete · PG00D=1 on battery.
- **5V_VDD** from the TPS61022 boost feeds the MAIN LDOs, backlight, MAX98357A and the WS2812s. No MCU enable — always on.
- USB-C lives entirely on the PWR board. **No USB on the STM32** — do not configure USB_OTG.

RECOMMENDED BOOT SEQUENCE main()

- 1 HAL_Init · MPU_Config · CPU_CACHE_Enable
 - 2 SystemClock_Config (VOS0 + ODEN + 4WS)
 - 3 MX_GPIO_Init – first of all. Guarantees PB3=0, PD8=0, PF11=0, PE13=0, PG1=0.
 - 4 MX_DMA_Init · MX_QUADSPI_Init → read JEDEC ID, set QE, enter memory-mapped mode
 - 5 MX_I2C2/3/4_Init · bus scan (log every address that answers)
 - 6 Display: PE13=1 → 10 ms → panel serial init via PG0/PA8/PC9 → MX_LTDC_Init → DMA2D → clear framebuffer → only then PG1=1
 - 7 IMU: WHO_AM_I (0x0F) must read 0x6A → CTRL3_C, ODR, INT1_CTRL
 - 8 ToF: PF2=1 → 1 ms → v15315cx_init() (slow firmware upload) → start ranging
 - 9 Audio: MX_SAI1_Init + MX_I2S3_Init, circular DMA, buffers zeroed
 - 10 Wi-Fi: PB3=0, PD8=0 → PF11=1 → 5 ms → PB3=1 → wait PB4=1 → read "ready" → send "AT" → expect "OK"
 - 11 LEDs: TIM1 PWM + DMA, first boot animation
 - 12 ADC calibration + first battery reading
- ORDER MATTERS** If you call MX_SPI2_Init() before PB3 is driven low, the module will already have powered itself up (R17 pulls CHIP_EN to 3V3_WIF1) and you will miss the "ready" message.

MEMORY BUDGET 1 MB RAM total

REGION	BASE	SIZE	USE
AXI SRAM (D1)	0x24000000	512K	RGB565 framebuffer 450K
SRAM1 (D2)	0x30000000	128K	audio DMA buffers
SRAM2 (D2)	0x30020000	128K	Wi-Fi / LwIP buffers
SRAM3 (D2)	0x30040000	32K	ToF / IMU
SRAM4 (D3)	0x38000000	64K	low-power / logs
DTCM	0x20000000	128K	stack + DSP hot path
ITCM	0x00000000	64K	critical ISRs
• Double buffering does not fit at 480×480. Use LVGL partial rendering (two ~40-line buffers) into the single framebuffer. • Never put DMA buffers in DTCM — DMA1/DMA2 cannot reach TCM. Use D2 (SRAM1/2).			

SHEET-TO-SHEET CONFLICTS resolved via netlist

SIGNAL	SHEET 7	SHEET 4 (MCU)	CORRECT
LCD_CS	PE10	PG0	PG0
LCD_BL	PE9	PG1	PG1
DATA_LED	PA8 (PWM)	PA9	PA9

PF9/PE10 are marked **NC** on sheet 4, and PA8 is already LCD_SCL. The

HARDWARE FINDINGS read before debugging firmware

- 1 **NO PULL-UPS ON ANY I2C BUS** I2C2 (ToF), I2C3 (display) and I2C4 (IMU) have no pull-up resistors on the MAIN board. Internal pull-ups work but lock you at 100 kHz. **Fit 2.2 kΩ per line**, or accept the slow ToF boot.
- 2 **IMU SAO AT AN UNDEFINED LEVEL** R23/R24 (33K each) form a divider into SDO/SAO → ~1.65 V, between V_{IL} (0.99 V) and V_{IH} (2.31 V). Address may resolve as 0x6A or 0x6B, and vary unit to unit. Remove R23 (forces 0x6A) or R24 (forces 0x6B); probe both in firmware.
- 3 **/PGOOD AND /CHG PULLED TO 5 V** R3/R4 (33K) pull to **5V_VDD**, but PC4 and PA7 are not 5 V-tolerant. The ~42 μA injected stays inside the ±5 mA limit, but injection degrades ADC accuracy on the same block — and BATT (PC5) is exactly there. Move the pull-ups to 3V3, or average heavily in firmware.
- 4 **MAX98357A CANNOT BE SHUT DOWN** R9 (100K → 5 V) + R8 (634K → GPIO) hold SD_MODE at 4.32 V even with PB6 low — always above the 1.4 V threshold. The amp is permanently on, left channel. Mute by stopping BCLK (self-shutdown in ~50 ms); change R8 to ≤ 3.3 kΩ in REV02.
- 5 **WS2812 LOGIC LEVEL** 5 V parts need V_{IH} = 3.5 V but PA9 gives 3.3 V. May work on the bench and fail over temperature.
- 6 **SAI_D2 (PE4) IS A DEAD NET** It leaves the MCU but the second mic pin carries a red X on sheet 9 — both mics sit on PC1. Design intent, not a firmware bug.

BRING-UP CHECKLIST in order

- ☐ TP1/TP2/TP3 read 5V_VDD / 3V3_MCU / GND before plugging in the ST-LINK
- ☐ SWD connects and reads DEVID (0x450)
- ☐ BLink on PG1 (backlight) confirms clock and GPIO
- ☐ HSE 25 MHz oscillates (probe the crystal pin; PA8 is not free for MC01)
- ☐ QSPI JEDEC ID reads 0xEF 0x40 0x16
- ☐ I2C scan: 0x29 (ToF) and 0x6A/0x6B (IMU) respond
- ☐ ISM330DLC WHO_AM_I = 0x6A
- ☐ LTDC colour bars stable, no tearing (check MPU/cache)
- ☐ SAI PDM buffer is neither constant nor all zeros
- ☐ I2S: a 1 kHz sine comes out of the speaker cleanly
- ☐ Wi-Fi: PB4 rises after PB3=1 and the module answers "ready"
- ☐ All 10 WS2812s show the right colour (LED1 wrong → logic level)
- ☐ Battery ADC matches the multimeter within ±50 mV
- ☐ Wi-Fi TX current does not brown out the boost